

- 2 Resistance to antibiotics within a population of bacteria is due to selection pressure. This can be linked to the use of antibiotics by patients.

A study was carried out into the link between antibiotic use and the presence of resistant *Escherichia coli* (*E. coli*) populations in human communities.

- Over 30 000 patients were involved in the study.
- Only patients attending large medical clinics took part in the study.
- The number of prescriptions issued by each clinic was used as an estimate of antibiotic use.
- Urine from patients attending the clinics was used as a possible source of antibiotic resistant *E. coli*.
- Antibiotic resistance of *E. coli* in the urine samples was measured using the disc diffusion method.

The disc diffusion method measures sensitivity of bacteria to an antibiotic. A bacterial population with low sensitivity to an antibiotic is resistant to that antibiotic.

In the disc diffusion method a Petri dish is filled with nutrient agar and urine samples containing *E. coli* are spread evenly across the agar.

Discs containing different antibiotics are placed on top of the agar. A lid is put on the Petri dish and the plate is incubated overnight.

Fig. 2.1 shows an example of a Petri dish from the study after incubation.

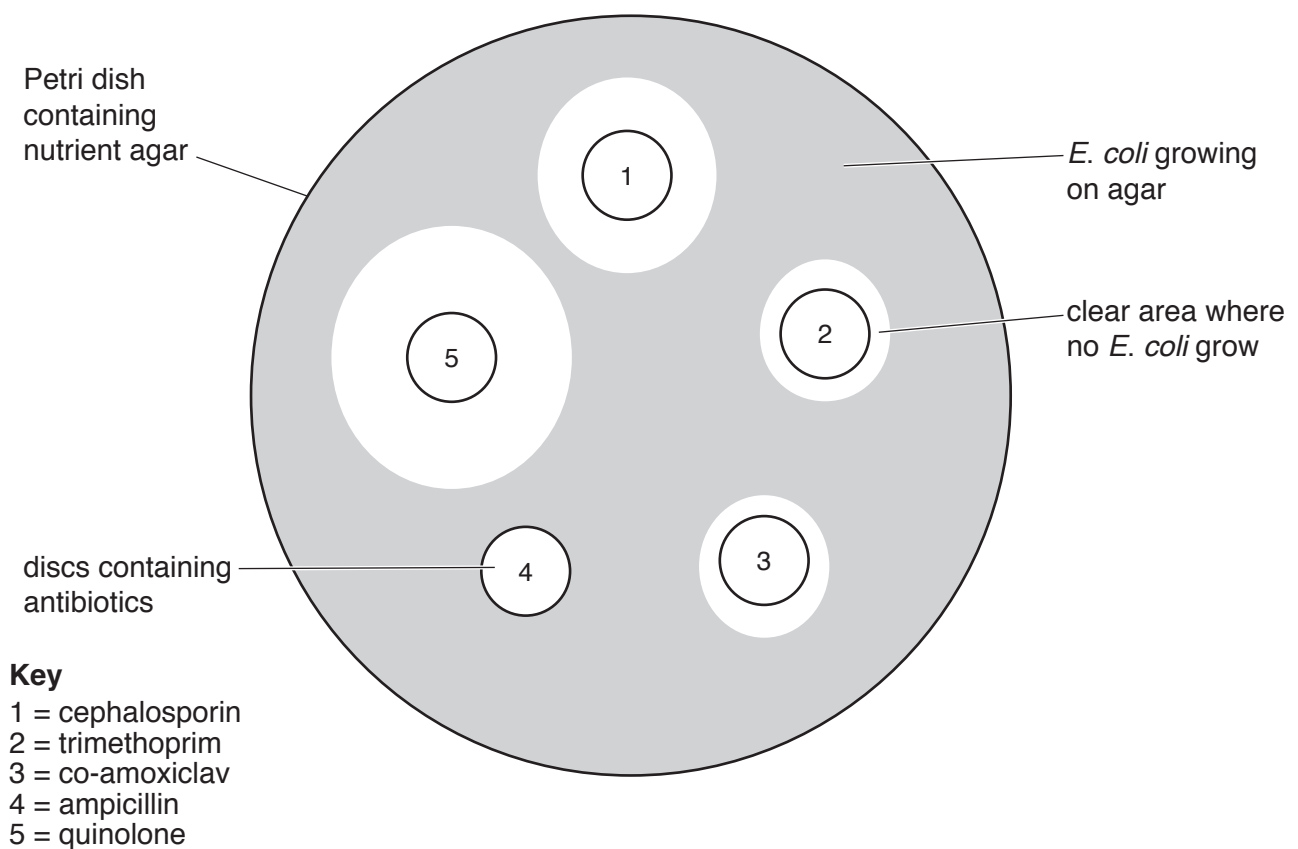


Fig. 2.1

- (a) (i) Suggest two variables that need to be standardised when using the disc diffusion method in this study.

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[2]

- (ii) Describe how you would determine the sensitivity of *E. coli* to each antibiotic.

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(b) Table 2.1 shows the results of this investigation.

Table 2.1

antibiotic	antibiotic use /prescriptions per thousand patients per year		percentage <i>E. coli</i> resistance	
	mean ($\bar{x}$)	standard deviation (s)	mean ($\bar{x}$)	standard deviation (s)
cephalosporin	107.0	83.0	6.5	3.5
trimethoprim	62.6	25.6	26.3	5.8
co-amoxiclav	75.5	43.9	8.4	5.7
ampicillin	351.9	171.1	53.2	7.2
quinolone	33.6	18.3	2.2	1.9

(i) Comment on the standard deviations for **antibiotic use** as shown in Table 2.1.

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[2]

(ii) Suggest two reasons why the number of prescriptions issued for antibiotics may **not** give an accurate measure of antibiotic use by patients.

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[2]

- (c) The number of prescriptions issued for antibiotics varied considerably between clinics.

The researchers wanted to find out whether there was a correlation between the number of prescriptions for each of the five antibiotics issued by a clinic and the percentage of urine samples containing resistant *E. coli*.

Spearman's rank correlation test was used for this analysis.

The results of this analysis are shown in Table 2.2.

Table 2.2

antibiotic	Spearman's rank correlation coefficient (r_s)
cephalosporin	0.30
trimethoprim	0.62
co-amoxiclav	0.23
ampicillin	0.71
quinolone	0.44

Table 2.3 shows the critical values for r_s at five levels of significance for the data collected in this study.

Table 2.3

level of significance (p)	0.20	0.10	0.05	0.02	0.01
critical value of r_s	0.240	0.306	0.362	0.425	0.467

- (i) Suggest why the Spearman's rank correlation test was used in this study.

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[1]

- (ii) State a null hypothesis for the Spearman's rank correlation test for this study.

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[1]

- (iii) Using Table 2.2 and Table 2.3, identify which antibiotics showed a statistically significant correlation between the number of prescriptions and the presence of resistant strains of *E. coli* in urine samples. Give a reason for your answer.

antibiotics

reason

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- (d) The percentage of patients with ampicillin-resistant *E. coli* infections varies with age and gender, as shown in Fig. 2.2.

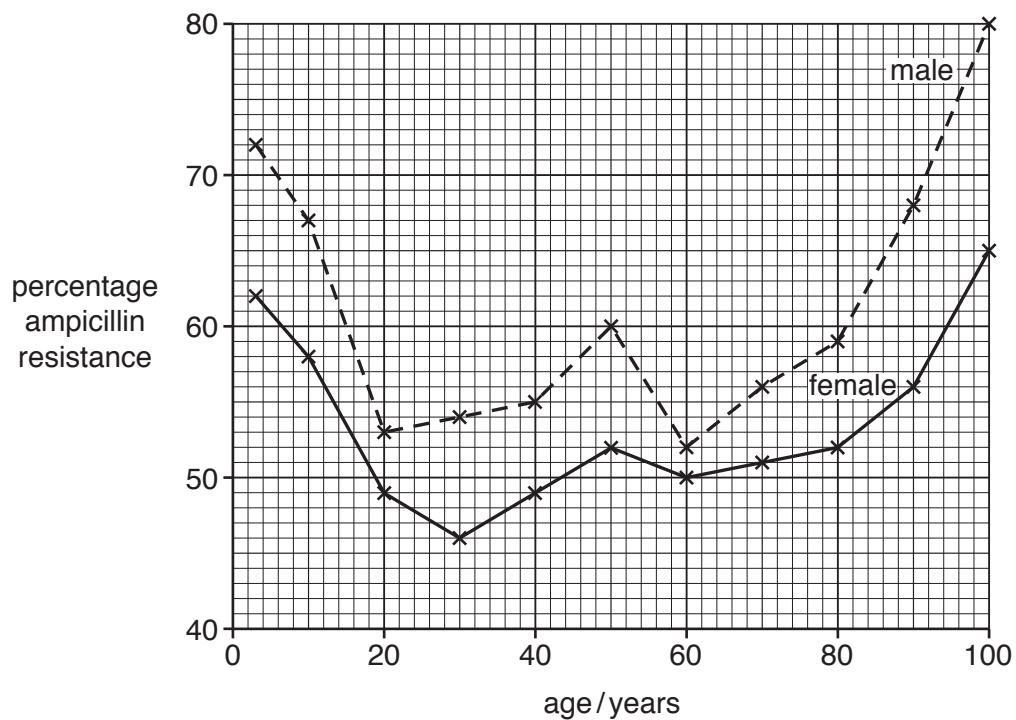


Fig. 2.2

Describe **and** explain the trends shown by these data.

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[Total: 15]